Name: ____________________

PROPORTION, (PROPORTION GRAPHS)

Date: ______________

LO: I can identify and interpret graphs showing direct and inverse proportion.

Proportion graphs

A **direct proportion** graph is a **straight line through the origin**. The gradient equals the constant k.

An **inverse proportion** graph is a **curve (hyperbola)** that gets closer to the axes but never touches them.

Recognising graphs

	y = 3x: straight line through (0,0)	direct proportion
	y = 12/x: curve in first quadrant	inverse proportion
	Steeper line = larger k	gradient = k
	y = 2x + 1 is direct proportion	does not pass through origin
	Inverse proportion is a	

Key idea

Direct: straight line through origin (y = kx). Inverse: hyperbola (y = k/x). Read k from the graph.

Direct: gradient = k | Inverse: xy = k

Both can be identified from their graph shape

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – identifying direct proportion

A graph passes through (0,0) and (4, 20). Is it direct proportion? Find k.

Passes through origin: yes, could be direct proportion

k = y/x = $\frac{20}{4}$ = 5

Formula: y = 5x

Yes, direct proportion with k = 5

A straight line through the origin confirms direct proportion.

Example 2 – reading an inverse graph

A curve passes through (2, 12) and (4, 6). Show this is inverse proportion.

Check: 2 × 12 = 24; 4 × 6 = 24

Product xy is constant = 24

Therefore y = 24/x (inverse proportion)

Inverse proportion: y = 24/x

If xy is constant for all points, it is inverse proportion.

Example 3 – from table to graph type

x: 1, 2, 3, 4. y: 5, 10, 15, 20. What type of proportion?

y/x = 5, 5, 5, 5 (constant ratio)

This means y = 5x

Direct proportion (line through origin with gradient 5)

Direct proportion, y = 5x

Constant ratio y/x confirms direct proportion.

Example 4 – from table - inverse

x: 1, 2, 4, 8. y: 40, 20, 10, 5. What type?

xy = 40, 40, 40, 40 (constant product)

This means y = 40/x

Inverse proportion

Inverse proportion, y = 40/x

Constant product xy confirms inverse proportion.

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying from description

State whether each describes direct or inverse proportion.

a) Straight line through origin

b) Curve approaching axes

c) y/x is constant

d) xy is constant

2. Finding k from graphs

Find k from the given point on a direct proportion graph (through origin).

a) (3, 15)

b) (4, 28)

c) (6, 9)

d) (10, 45)

3. Inverse proportion from points

Verify inverse proportion and find k.

a) (2, 18) and (6, 6)

b) (3, 20) and (5, 12)

c) (4, 10) and (8, 5)

d) (1, 50) and (10, 5)

4. Tables to proportion type

State the type of proportion and find k.

a) x: 2,4,6 y: 8,16,24

b) x: 2,5,10 y: 30,12,6

c) x: 3,6,9 y: 12,24,36

d) x: 4,8,12 y: 15,7.5,5

5. Mixed graph problems

Identify and use the proportion relationship.

a) Line through (0,0) and (5, 30). Find y when x = 8.

b) Curve through (3, 24). Find y when x = 6 (inverse).

c) Is y = 4x + 2 direct proportion? Explain.

d) Graph through (2, 50). If inverse, find y at x = 10.

e) Line gradient 7 through origin. Find x when y = 42.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) A direct proportion graph is a straight line through the origin ☐

Correct answer: __________

Reason:

b) y = 3x + 2 is a direct proportion graph ☐

Correct answer: __________

Reason: ________________

c) An inverse proportion graph is a hyperbola ☐

Correct answer: __________

Reason: ________________

d) Any straight line shows direct proportion ☐

Correct answer: __________

Reason: ________________

e) The gradient of a direct proportion graph equals k ☐

Correct answer: __________

Reason: ________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A graph passes through the points **(2, 8), (4, 4)** and **(8, 2)**. (i) Show this is inverse proportion. (ii) Find k. (iii) Predict y when x = 16.

Expression: ____________________________

Simplified: ____________________________

b) Two variables are tested. x: 1, 2, 3, 4, 5 and y: 3, 12, 27, 48, 75. Is this direct proportion? If not, investigate whether y x^2 or another relationship.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!